Dual-Rotor Dynamics in Double Spacetime Theory: Written Action, Free Mismatch, and a Neighbouring Oscillator

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Abstract

In double spacetime theory every massive particle carries an intrinsic pair of usual and dual Minkowski sectors, encoded in the 16-real-dimensional biquaternion algebra isomorphic to $\mathrm{Cl}(3,1)$. Gravity is the torsional mismatch of the usual rotor R_{usual} and the dual rotor R_{dual}. A particle-intrinsic action for the six rapidities is a model, not a deduction from unitarity of those rotors. Opposite kinetic signs become a mixed term in the common rapidity $\sigma = \phi + \theta$ and the mismatch $\delta = \phi - \theta$. The mismatch is free, while the mass sources the common rapidity: $\ddot{\sigma} = -2m\delta$. Affine lag integrates to a cubic common rapidity, and along that motion the Jacobi integral reduces to initial data. The integral is indefinite, so conservation does not restore a frequency. A neighbouring sign pattern, sometimes written alongside the action, integrates to a hyperbolic runaway at the same rate $\sqrt{2m}$. An oscillator appears only when both kinetic signs are taken positive, and then at frequency $\sqrt{2m}$, which meets the Compton frequency m only at the isolated values $m = 0$ and $m = 2$. Linear accumulation of free lag is the only secular phase the written action supplies; identifying that rate with the de Broglie phase remains a reading. The oscillator potential of this note is not the parameter-space Killing quadratic of the main paper. For every potential of the mismatch the opposite-sign lag stays free, so the harmonic stiffness accelerates a nonzero lag; a well off alignment freezes a lag only when the kinetic signs agree. On one axis that critical alignment is not the vacuum: $J = \frac{1}{2}\delta\sigma$ and $M = \frac{1}{4}(\sigma^2 + \delta^2)$, so $\delta = 0$ leaves $M = \sigma^2/4$. On one axis the same coordinates split a null translation into a plane scaled by the usual rapidity and a complementary plane rotated by the dual angle. The mass accelerates both and does not restore their offset. Off that axis a usual generator and a dual generator anticommute, and the sign of $p^2 - q^2$ selects one hyperbolic, elliptic, or parabolic rotor rather than two planes. The parabolic rotor is a null rotation of the Minkowski frame. Its truncations form a one-parameter group of isometries, fixing one null line, and the same coefficients act on translations. A translation has the same truncated exponential and displaces a hyperplane along e_4.

1 Introduction

The double spacetime theory refuses a shared continuum. Each massive particle carries two compactified Minkowski frames inside the 16-real-dimensional biquaternion algebra $\mathbb{H}\otimes\mathbb{H} \cong \mathrm{Cl}(3,1)$ [1]. The complete rotor is written

$$R_{\mathrm{total}} = \exp\left[\sum_{a=1}^{3}\left(\frac{\phi_a}{2}\, i\Gamma_a + \frac{\theta_a}{2}\,\Gamma_a\right)\right], \qquad \Gamma_1 = I,\ \Gamma_2 = J,\ \Gamma_3 = K.$$

The hyperbolic generators $i\Gamma_a$ square to $+1$ and move the usual sector; the cyclic generators Γ_a square to -1 and rotate the dual sector. Same-axis usual and dual generators commute, so a one-axis exponential factorises. Distinct-axis generators anticommute, and $p\, i\Gamma_a + q\,\Gamma_b$ squares to the scalar $p^2 - q^2$. An unrestricted six-parameter factorisation $R_{\mathrm{usual}}R_{\mathrm{dual}}$ is not assumed here.

The relative rotor $\Omega = R_{\mathrm{usual}}^{\dagger}R_{\mathrm{dual}}$ is the identity if and only if the two rotors coincide. A Lorentz-invariant scalar J built from the mismatch bivector $\Omega_{\mathrm{biv}} = \log\Omega$ measures that disagreement in the classical field theory. The J of this note is a different object: a small-angle oscillator potential assembled from the mismatch angles of a single particle. The admissible ceiling $|J_{\mathrm{norm}}| \le 1$ of the main paper must not be transplanted here.

The question is dynamical. Once the mismatch angle $\delta_a = \phi_a - \theta_a$ is promoted to a coordinate of a particle-intrinsic Lagrangian, what motion follows for δ and for the common rapidity $\sigma = \phi + \theta$? Three neighbouring systems have been compared. Their Euler–Lagrange identities are elementary, and they do

not agree. The geometry of the dual rotor—its real character, its spinor matrix element, its SU(2) pairing with an observer—belongs to a companion note [2]. What is asked here is the motion of that pair.

2 Torsional mismatch as a particle potential

At an equal-rapidity state a pure lag δ adds $(m/2)\delta^2$ to the potential and $\delta^2/4$ to the unsigned mass. The common acceleration per unit lag is minus the ratio of those two coefficients of δ^2. This note does not fix the value of m. The oscillator potential

$$J_{\text{osc}} = \frac{1}{2} m \sum_{a=1}^{3} \delta_a^2$$

is not the parameter-space Killing quadratic $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ of the main paper. Vanishing of that Killing scalar does not force $\Omega = 1$; a balanced massive seed may have $J = 0$ with a nontrivial relative rotor. On one axis the relative rotor is not a function of the mismatch alone: its first-order part retains the common rapidity. The potential J_{osc} knows nothing of either distinction. It is quadratic in the coordinate difference δ_a and nothing else.

3 The written particle action

The written effective action for the intrinsic angles of a single particle is

$$S_{\text{particle}} = \int \left[\frac{1}{2} \sum_{a=1}^{3} \dot{\phi}_a^2 - \frac{1}{2} \sum_{a=1}^{3} \dot{\theta}_a^2 - \frac{m}{2} \sum_{a=1}^{3} (\phi_a - \theta_a)^2 \right] dt.$$

The opposite signs of the kinetic terms are part of the written model. They are intended to reflect boost freedom in the usual sector and the time-reversed, elliptic character of the dual sector. They are not deduced from unitarity of the rotors.

On one axis the density is

$$L = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\dot{\theta}^2 - \frac{m}{2}(\phi - \theta)^2.$$

The Euler–Lagrange equations of this density are

$$\ddot{\phi} = -m(\phi - \theta), \qquad \ddot{\theta} = -m(\phi - \theta). \tag{1}$$

Both angles accelerate by the same amount. The mismatch $\delta = \phi - \theta$ therefore satisfies

$$\ddot{\delta} = 0. \tag{2}$$

The same identities are transparent in the sum and the difference. Write $\sigma = \phi + \theta$. Opposite kinetic signs are then a mixed term,

$$L = \frac{1}{2}\dot{\sigma}\dot{\delta} - \frac{m}{2}\delta^2. \tag{3}$$

The coordinate σ does not appear in the density. Its conjugate momentum $\frac{1}{2}\dot{\delta}$ is therefore conserved, which is (2) again. What the potential actually sources is the common rapidity,

$$\ddot{\sigma} = -2m\delta. \tag{4}$$

The free mismatch integrates at once. On one axis its general solution is the affine lag

$$\delta(t) = \delta_0 + \nu t, \tag{5}$$

where the rate ν is initial data, not a mass. Twice integrating (4) fixes the common rapidity,

$$\sigma(t) = \sigma_0 + \dot{\sigma}_0\, t - m\delta_0\, t^2 - \frac{m\nu}{3}\, t^3. \tag{6}$$

A frozen lag, $\nu = 0$, accelerates σ uniformly. A nonzero rate makes the acceleration itself linear in time, and the common rapidity cubic. The mass m, written as a stiffness of δ, is a force on σ.

The Jacobi integral

$$E = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\dot{\theta}^2 + \frac{m}{2}\delta^2 = \frac{1}{2}\dot{\sigma}\dot{\delta} + \frac{m}{2}\delta^2$$

is conserved along (1). On the cubic flow it collapses further, to a constant fixed entirely by the initial data,

$$E = \frac{1}{2}\nu\dot{\sigma}_0 + \frac{m}{2}\delta_0^2.$$

The quadratic growth of $\dot{\sigma}$ and the quadratic growth of the potential cancel, so the charge stays finite while the rapidity runs away. Mixed kinetics still make E indefinite: opposing signs of ν and $\dot{\sigma}_0$ yield $E < 0$, as does a purely dual velocity at vanishing lag. Conservation of that constant does not select a frequency.

Opposite kinetic signs, which were meant to distinguish the two sectors, cancel in the mismatch channel and survive only as a drive of the common rapidity. What remains of the potential in (1) is a common force, not a restoring force on δ.

Collective averaging of this particle-intrinsic action over many particles is sometimes proposed as recovering a macroscopic integral of J. That averaging is a reading. It is not an identification of J_{osc} with the Weitzenböck density T.

4 Two planes of a null translation

The rapidities are coefficients of bivectors. A null vector e_4, adjoined to the Minkowski frame, makes a translation a null bivector $N_\mu = e_4 \wedge e_\mu$. On one axis the usual generator is the boost in one plane, and the dual generator is the rotation in the complementary spatial plane. They commute, the exponential factorises, and each stabilises the other's plane.

A null translation therefore splits. In the boost plane the factorised rotor scales the lightlike combinations by the usual rapidity alone,

$$N_0 + N_1 \mapsto e^{\phi}(N_0 + N_1), \qquad N_0 - N_1 \mapsto e^{-\phi}(N_0 - N_1).$$

In the complementary plane it rotates by the dual angle alone,

$$N_2 \mapsto \cos\theta\, N_2 + \sin\theta\, N_3, \qquad N_3 \mapsto -\sin\theta\, N_2 + \cos\theta\, N_3.$$

The common rapidity $\sigma = \phi + \theta$ dresses both planes by the same function, and the mismatch $\delta = \phi - \theta$ is the offset between them. The plane parameters are $(\sigma + \delta)/2$ and $(\sigma - \delta)/2$.

Along the written flow that offset is the affine lag, and the common dressing is the cubic rapidity. Each plane has second derivative $-m\delta$. The stiffness accelerates both planes together, and their difference remains free, so a nonzero lag is not restored. A frozen lag is a constant separation of the hyperbolic eigenvalue from the elliptic angle, while both run with σ.

The relative rotor acts by the inverse boost $-\phi$ on the first plane and by the rotation θ on the second. Neither action is a function of δ alone. The pairs $(\phi, \theta) = (0, 0)$ and $(1, 1)$ share a vanishing mismatch and move the two planes differently. At the identity the first-order change along (ϕ, θ) is $-\frac{\phi}{2}\, i\Gamma + \frac{\theta}{2}\, \Gamma$, where $i\Gamma$ and Γ are the usual and dual generators of that axis. In the sum and the difference,

$$-\frac{\sigma}{4}\,(i\Gamma - \Gamma) - \frac{\delta}{4}\,(i\Gamma + \Gamma).$$

A chart that retains only $\delta(i\Gamma + \Gamma)$ drops the common rapidity. Alignment $\delta = 0$ with $\sigma \neq 0$ still scales the boost plane by $e^{\sigma/2}$.

Off that axis the split stops. A usual generator and a dual generator on distinct axes anticommute, so a combination $p\, i\Gamma_a + q\, \Gamma_b$ squares to the scalar $p^2 - q^2$. The sign of that scalar is the type of a single rotor. When $p^2 - q^2 = s^2 > 0$,

$$\exp(p\, i\Gamma_a + q\, \Gamma_b) = \cosh s + \frac{\sinh s}{s}\,(p\, i\Gamma_a + q\, \Gamma_b),$$

and when $q^2 - p^2 = s^2 > 0$ the same formula holds with cos and sin. When $p^2 = q^2$ the exponential truncates at first order,

$$\exp(p\, i\Gamma_a + q\, \Gamma_b) = 1 + p\, i\Gamma_a + q\, \Gamma_b.$$

None of the three is a product of a boost and a rotation. Whenever $\sinh p \sin q \neq 0$, the ordered product carries an extra multiple of the remaining boost.

The generator $i\Gamma_0 + \Gamma_1$ lies in the Lorentz span, meets the null ideal only at zero, and squares to zero. The truncated rotor $1 + t(i\Gamma_0 + \Gamma_1)$ is a null rotation of the Minkowski frame. It fixes the null line $e_0 - e_3$ and the transverse axis e_2, and it shears the remaining frame,

$$e_1 \mapsto e_1 + 2t(e_0 - e_3), \qquad e_0 + e_3 \mapsto (e_0 + e_3) + 4t\, e_1 + 4t^2(e_0 - e_3). \tag{7}$$

The cube of the adjoint action vanishes on vectors, so the dependence on t stops at the quadratic term. The truncations form a group under that nilpotence: the product at parameters t and s is the truncation at $t + s$, and $-t$ is the inverse. Reversal sends each truncation to that inverse, so the shear preserves the Minkowski product. For $t \neq 0$ the vectors fixed by the shear are the span of $e_0 - e_3$ and e_2, and these two directions are orthogonal.

The null vector e_4 commutes with the Lorentz span, so the translational ideal carries the same coefficients. The rotor sends $N(x) = \sum_\mu x^\mu N_\mu$ to $N(x')$, where x' is the image of x under (7). Thus N_1 goes to $N_1 + 2t(N_0 - N_3)$ and leaves the boost plane when $t \neq 0$, while a lightlike translation remains lightlike. A translator $1 + sN_1$, of the same truncated shape, sends the hyperplane e_1 to $e_1 + se_4$.

5 Two neighbouring systems

A neighbouring system is sometimes written alongside the action, as if it were the Euler–Lagrange system of the same Lagrangian:

$$\ddot{\phi} = m(\phi - \theta), \qquad -\ddot{\theta} = m(\phi - \theta). \tag{8}$$

This is not the Euler–Lagrange system of the written density. The mismatch then runs away,

$$\ddot{\delta} = 2m\delta. \tag{9}$$

It is not an oscillator. The common rapidity is free: $\ddot{\sigma} = 0$. For $m > 0$, with rate $\kappa = \sqrt{2m}$, the lag integrates to the hyperbolic motion

$$\delta(t) = A\cosh(\kappa t) + B\sinh(\kappa t),$$

while σ stays inertial. A nonzero seed is unbounded in time. The mass, instead of restoring the lag, amplifies it.

An oscillator does appear if both kinetic signs are taken positive. The density

$$L = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\dot{\theta}^2 - \frac{m}{2}(\phi - \theta)^2$$

has Euler–Lagrange equations

$$\ddot{\phi} = -m(\phi - \theta), \qquad \ddot{\theta} = m(\phi - \theta),$$

and therefore

$$\ddot{\delta} + 2m\delta = 0. \tag{10}$$

The common rapidity is again free. This is an explicit working model, not the written action. In units $\hbar = c = 1$ its angular frequency is $\sqrt{2m}$. That line and the Compton frequency m are different functions of the rest mass: they meet only at the isolated values $m = 0$ and $m = 2$. The Jacobi integral of this density is $\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\dot{\theta}^2 + \frac{m}{2}\delta^2$; for $m \geq 0$ it is nonnegative, so here conservation does bound the motion of δ. The general solution is an oscillation of the mismatch together with uniform common rapidity,

$$\delta(t) = A\cos\bigl(\sqrt{2m}\,t\bigr) + B\sin\bigl(\sqrt{2m}\,t\bigr),$$

the same family as a phase-shifted sine. The squared mismatch never exceeds the constant amplitude $A^2 + B^2$, and the Jacobi integral equals $\frac{1}{4}\dot{\sigma}_0^2 + \frac{m}{2}(A^2 + B^2)$ at every time. The three systems are neighbours in the space of quadratic Lagrangians. Kinetic signature—same versus opposite—decides whether δ is a dynamical coordinate. Opposite kinetics free the mismatch and let the potential drive σ. Same-sign kinetics free σ and leave the potential sign to decide oscillator versus runaway. Only one of the three is the written action, and that one does not oscillate. Only one of them oscillates, and that one is not the written action. The remaining system is runaway.

6 A dual-channel force

Gravitational engineering asks what happens if a dual-channel generalised acceleration u is added to $\ddot{\theta}$. On the written action the channels become

$$\ddot{\delta} = -u, \qquad \ddot{\sigma} = -2m\delta + u. \tag{11}$$

The mismatch remains free of a restoring frequency: it is the double integral of $-u$. Those identities assume both channels free. Holding $\ddot{\phi} = 0$ is a constraint. With a usual-channel force v admitted together with u, the written Jacobi integral changes at $v\dot{\phi} - u\dot{\theta}$, and $\ddot{\phi} = 0$ uniquely fixes $v = m(\phi - \theta)$. Dual-only kinematics is therefore not a free solution of the written action. At instantaneous usual rest the constraint contributes no power. On the same-sign working model,

$$\ddot{\delta} + 2m\delta = -u, \qquad \ddot{\sigma} = u. \tag{12}$$

A constant drive shifts the equilibrium mismatch by $-u/(2m)$ (when $m \neq 0$) while the common rapidity accelerates. Resonance is therefore additional structure, not an identity of either quadratic density. Identifying u with a laboratory Faraday bilinear is a modelling choice, not a derivation of a coupling to the particle mismatch.

7 Rigidity is a critical lag

The harmonic term in the written action is often read as the rigidity of a massive particle, and that rigidity as a geometric stand-in for the Higgs mechanism. The two readings separate as soon as the potential is an arbitrary function V of the mismatch.

Opposite kinetic signs do not consult the shape of V. The mismatch stays free, and the slope drives only the common rapidity:

$$\ddot{\delta} = 0, \qquad \ddot{\sigma} = -2V'(\delta).$$

A constant lag is inertial only where the slope vanishes. The harmonic stiffness $V = \frac{m}{2}\delta^2$ vanishes only at alignment, so a nonzero lag, with $m \neq 0$, accelerates σ. The mass term does not select a misalignment.

Alignment is not the vacuum, the state in which every rapidity vanishes. On one axis the usual and dual rapidities are $(\sigma + \delta)/2$ and $(\sigma - \delta)/2$, and

$$J = \frac{1}{2}\delta\sigma, \qquad M = \frac{1}{4}(\sigma^2 + \delta^2).$$

The critical point $\delta = 0$ sets $J = 0$ and leaves $M = \sigma^2/4$. The written flow keeps that alignment when the lag rate vanishes, and then σ coasts. M is constant in time if and only if the coasting rate vanishes. A nonzero rate meets $M = 0$ once; away from that instant the two rotors still disagree, so the passage is not free fall. The vacuum is a crossing of an inertial line.

Same-sign kinetics exchange the roles. The common rapidity is free for every V, and the mismatch feels the slope,

$$\ddot{\delta} = -2V'(\delta), \qquad \ddot{\sigma} = 0.$$

Constant lag and vanishing slope are then the same condition, so the harmonic well still admits only alignment. The quartic

$$V(\delta) = \frac{\lambda}{4}(\delta^2 - v^2)^2$$

is minimized, for $\lambda > 0$, at $\delta = \pm v$, where it vanishes, and the symmetric point lies strictly above those wells when $v \neq 0$. Between the wells the same-sign force has the sign of δ, so the origin repels. At either well the lag may freeze, while the common rapidity stays inertial at a rate fixed by initial data. A small departure has curvature $4\lambda v^2$. That number can be set equal to a Compton value m^2, and it can miss it.

Along a frozen well the common rapidity still coasts, and both scalars travel with it:

$$J = \frac{1}{2}v\sigma, \qquad M = \frac{1}{4}(\sigma^2 + v^2).$$

J is constant in time if and only if that coasting rate vanishes. At rest, with $\sigma = 0$ and $v \neq 0$, the well is a balanced massive state of mass $v^2/4$, and the rotors do not coincide. Under the written opposite signs an affine lag of nonzero rate meets every critical value once and does not stop there. A frozen nonzero lag, a massive radial mode, and a flat common phase are as much of a Higgs reading as one real potential of a single mismatch can bear. The flat direction is not gauged. No Weinberg angle appears, and W and Z receive no mass. The harmonic stiffness of the written action does not select the lag.

8 Secular phase and the de Broglie identification

The written action does not furnish a harmonic bridge to quantum theory. What it does furnish is linear accumulation of free mismatch, together with a polynomial common rapidity sourced by that lag. That common motion is not a wave. If the initial rate ν is read as proportional to the laboratory four-momentum, the accumulated lag

$$\int \delta \, dt \propto \frac{Et - \mathbf{p} \cdot \mathbf{x}}{\hbar}$$

is the de Broglie phase of the matter wave [3]. That identification remains a reading. The mass that appears in the potential does not select the rate; it accelerates σ. The coefficient ν is data.

If instead the same-sign model is adopted as the dynamics, the common rapidity is uniform, and rapid oscillations at $\sqrt{2m}$ must still be averaged by hand before a slow secular phase can be claimed. The line, if it is physical, sits at $\sqrt{2m}$. It meets m only at the isolated values $m = 0$ and $m = 2$. Absence of a narrow resonance at either frequency would constrain the dynamics, not the dual-rotor kinematics.

The Compton wavelength continues to be readable as a compactification radius of the dual sector, $\sim \hbar/mc$. The de Broglie wavelength is the laboratory distance over which an identified scalar phase advances by 2π. Neither length is derived from the oscillator, and neither is the trace of a dual rotor.

The scalar U(1) factor $e^{i\delta}$ built from the mismatch angle is a third quantity, distinct from the real character $\operatorname{Tr} R(\beta) = 2\cos(\|\beta\|/2)$ of the dual rotor and from the complex spinor matrix element of that rotor [2]. Those two invariants belong to the SU(2) kinematics of the dual sector. They are not supplied by the particle action of this note, and the action does not turn them into a laboratory wave.

9 Conclusion

The dual-rotor pair is the geometry of a massive particle in double spacetime theory. The particle-intrinsic action is a model placed on that geometry, not a deduction from it. Of the three neighbouring systems compared here, the written opposite-sign density makes the mismatch free and sources the common rapidity: $\ddot{\delta} = 0$, $\ddot{\sigma} = -2m\delta$. The integral is the affine lag (5) together with the cubic rapidity (6), and the Jacobi integral along that motion is the constant $\frac{1}{2}\nu\dot{\sigma}_0 + \frac{m}{2}\delta_0^2$, conserved and indefinite. On one axis those two functions are the split of a null translation: $e^{\pm\phi}$ scales one plane and θ rotates the other, so the mass accelerates both planes and leaves their offset free. Off that axis the same two generators anticommute, and $p^2 - q^2$ selects one rotor—hyperbolic, elliptic, or parabolic—rather than two planes. The parabolic rotor is the null rotation (7), a one-parameter group of Minkowski isometries that fixes one null line and acts by the same coefficients on the translational ideal. A translator of the same truncated shape displaces a hyperplane along e_4. Same-sign kinetics free the common rapidity and leave the potential sign to decide oscillator versus runaway. At the common rate $\sqrt{2m}$ the runaway is hyperbolic and unbounded, while the oscillator is bounded by the squared amplitude $A^2 + B^2$. That oscillator frequency meets the Compton frequency of the rest mass only at the isolated values $m = 0$ and $m = 2$. A dual-channel force u freely accelerates the written mismatch, $\ddot{\delta} = -u$, and only shifts the oscillator's equilibrium. Holding the usual rapidity fixed against the potential requires a unique constraint force $v = m(\phi - \theta)$. That force pairs with $\dot{\phi}$ in the written energy balance; at instantaneous usual rest it is silent.

Identifying the free lag with the de Broglie phase, averaging J_{osc} over a many-particle ensemble, and reading the Compton wavelength as a dual-sector radius remain geometric readings. They are available. They are not forced by the Euler–Lagrange identities of the written action. The harmonic stiffness does not select a misalignment. Its critical alignment keeps $J = 0$ while M follows a coasting common rapidity, and a frozen well at rest is a balanced massive state. Neither is the vacuum.

References

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[2] Anonymous, "De Broglie Waves as Dual-Rotor Phase Lag: Quantum Mechanics as Geometry of Particle-Intrinsic Dual Spacetime" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/dst-de-broglie-wave.pdf`.

[3] L. de Broglie, "Recherches sur la théorie des quanta," *Ann. Phys.* (Paris) **3**, 22–128 (1925).